stelmec

beyond boundaries

POWER PRODUCTS | EPC | SMART GRID & POWER IT

Distribution Utility Meet- DUM26,
27th & 28th October 2026



Scaling Digital Rural Substation & Advancing Product Technology

Presented by RS Selvam , VP R&D

Stelmec Ltd. — Manufacturer of 11kV & 33kV Indoor/Outdoor VCB Panels, Kiosks & RMUs

TAKE CHARGE. THINK SMART.

Presentation Overview



1. Company Profile & Products
2. VI Manufacturing & Type Testing
3. VCB & VI Technology
4. IoT Substation Automation
5. AI in Substation Automation
6. IoT Case Studies
7. Conclusion
8. Discussion

1.0 About Stelmec

- **Four decades of experience in manufacturing, testing and supplying electrical power products.**
- **Customers include state utilities, EPC contractors and private industries in India and international markets.**
- **More than 25 years of indigenous MV switchgear development.**
- **Over 1,100 professionals and an installed base of more than 250,000 VCBs in India.**
- **Indoor and outdoor MV switchgear, including RMUs, supported by modern manufacturing facilities.**

1.0 Our Journey

1984 — Started as LV Switchboards Manufacturer



1990 — ‘SAFELINE’ Launch of MCBs under license of F&G Germany



1995 — Enter into JV with L&G-Switzerland



2000 — Incorporated in year 2000 as “STELMEC LIMITED”

2003 — Indigenously developed VCB & Panel for MV Segment



2012 — Launch of Relays under Stelmec brand



2016 — Incorporate EPC division, cater to T&D projects



2018 — Launch of Vacuum Interrupter under Stelmec



2024 — Launch of SF6 Ring Main Unit



1. 0 Switchgear- Introduction

Switchgear combines circuit breakers, switches, instrument transformers (CT/PT), protective relays, metering, fuses, surge arresters and control accessories — working together to isolate faults, control power flow and protect equipment and personnel in the network.

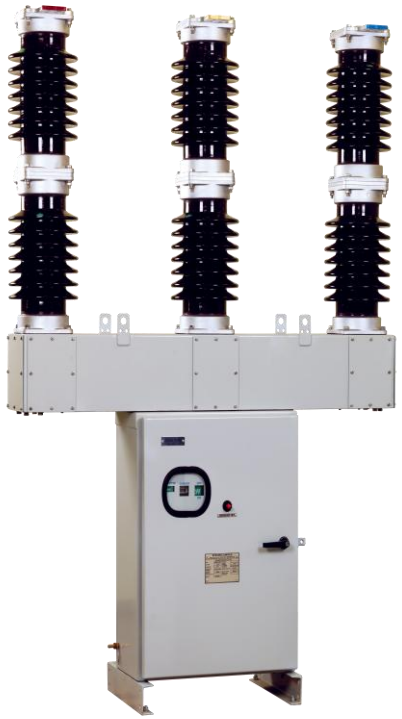
Medium voltage switchgear comes in various types based on insulation, construction, and function, including air-insulated, gas-insulated, solid-insulated, oil-insulated, metal-clad, metal-enclosed, compact, and pad-mounted designs.

MV switchgear is used in substations, factories, commercial buildings, power plants, renewable-energy facilities, and utility distribution networks

1.0 Stelmec Switchgear products family



36kV AIS



PCVCB



KIOSK



CRP



12kV AIS



12kV GIS



36 KV VCB TRUCK

1.0 Stelmec Switchgear products family

stelmec®
beyond boundaries
POWER PRODUCTS | EPC | SMART GRID & POWER IT

12kV VCB



36 KV VCB Truck



12/36 KV Kiosk



12KV RMU



36 KV outdoor VCB panel



Unit – I: Ularia, Ahmedabad









Unit – I: Ularia, Ahmedabad









Unit – II: Usgaon, Mumbai









Unit – II: Usgaon, Mumbai









Unit – II: Usgaon, Mumbai









Unit – III: Sativali, Mumbai









2.0 Advantage of VI Manufacturing

Many Rating of VCB are Type Tested for E2C2M2 such as:

- **12 & 36 KV VCB Kiosk Type Tested for E2C2M2**
- **12 & 36 KV PCVCB also Type Tested E2C2M2**
- **36 KV 31.5 KA 2500A Type Tested for E2C2M2 for 50/60Hz.**
- **12 & 36 KV Indoor panel also Type Tested from E2C2M2**
- **12 KV 40 KA 3150A also Under Test of E2C2M2.**

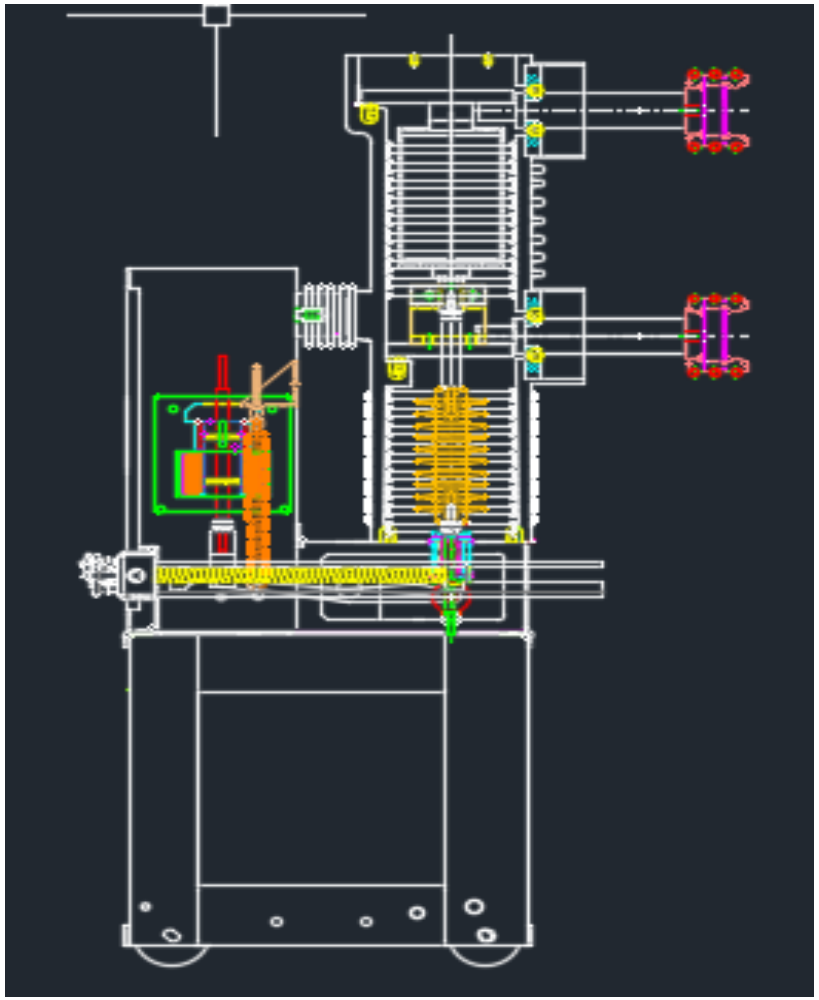
2.0 Advantage of VI Manufacturing

- **Solar-duty VCB under type testing.**
- **Rating: 36 kV, 31.5 kA, 3,150 A.**
- **Motorised mechanism with a dedicated VI.**
- **Target classification: E2C2M2.**



2.0 Advantage of VI Manufacturing

- **Furnace-duty VCB under development.**
- **Rating: 36 kV, 31.5 kA, 2,500 A.**
- **Magnetic actuator with a dedicated VI.**
- **Development target: 100,000 operations.**



2a. Classification of Switchgear

S1 class: Circuit-breaker intended to be used in a cable system.
S2 class: Circuit-breaker used in a line-system.
E1 class: Circuit-breaker with basic electrical endurance.
E2 class: Circuit-breaker with extended electrical endurance.(Auto reclosure & Non auto reclosure)
M1 class : Circuit-breaker with normal mechanical endurance (mechanically type tested for 2 000 operations).
M2 class: Circuit-breaker with extended mechanical endurance,(mechanically type tested for 10 000 operations.
C1 class: Circuit-breaker with low probability of re-strike during capacitive current breaking as demonstrated by specific type tests.
C2 class: Circuit-breaker with very low probability of re-strike during capacitive current breaking as demonstrated by specific type tests.

2a. Important of Periodic Type Tests

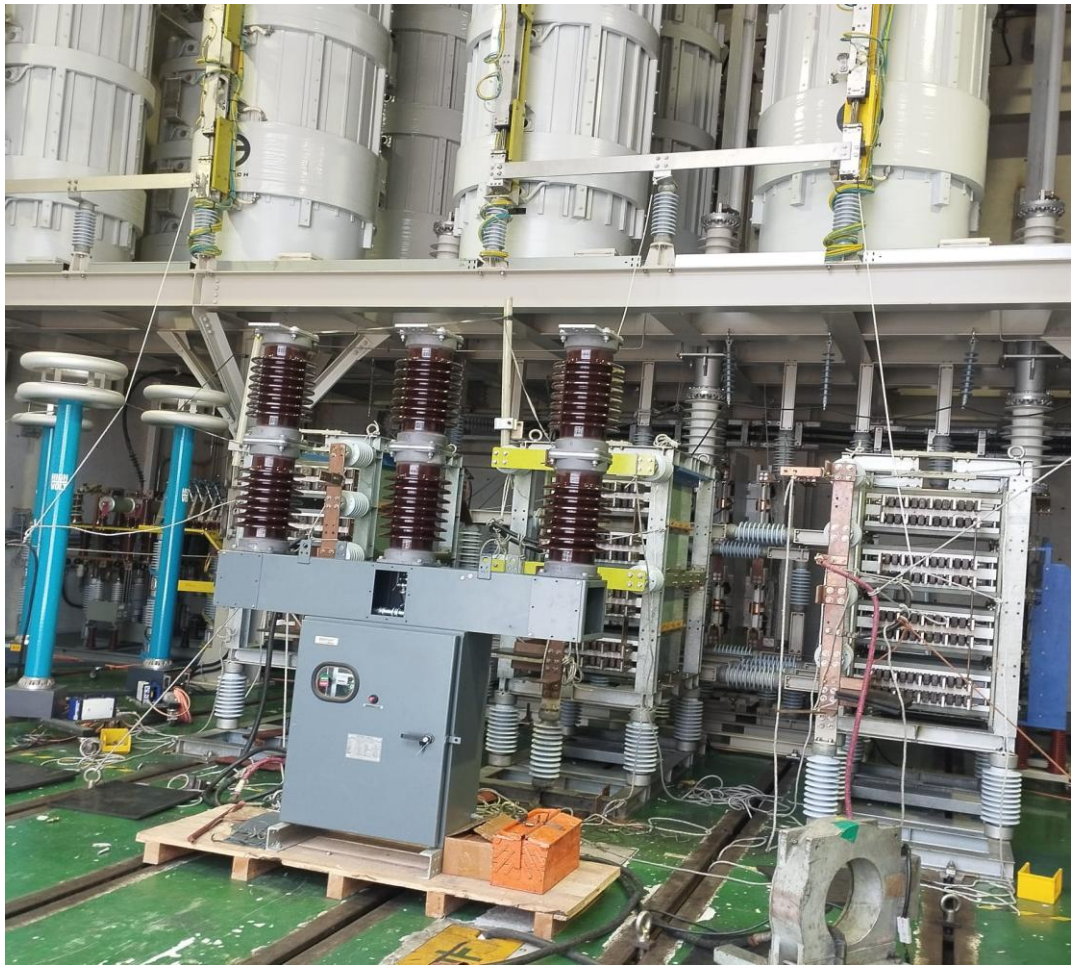
Necessity of Periodic Type Testing like 5 years duration

- As IEC standard is been revised every 3 to 4 years and hence it is to be verified and validated
- Also, if there is product Type Test upgradation Like EC2M2 instead of E1C1M1 for better performance at site.
- Enhancement of product performance though Design change having customer feedbacks & Site complaints ,etc.
- Any other Technology adaption etc.
- Repeated Design Verification & Validation

2a. Details Of Tests

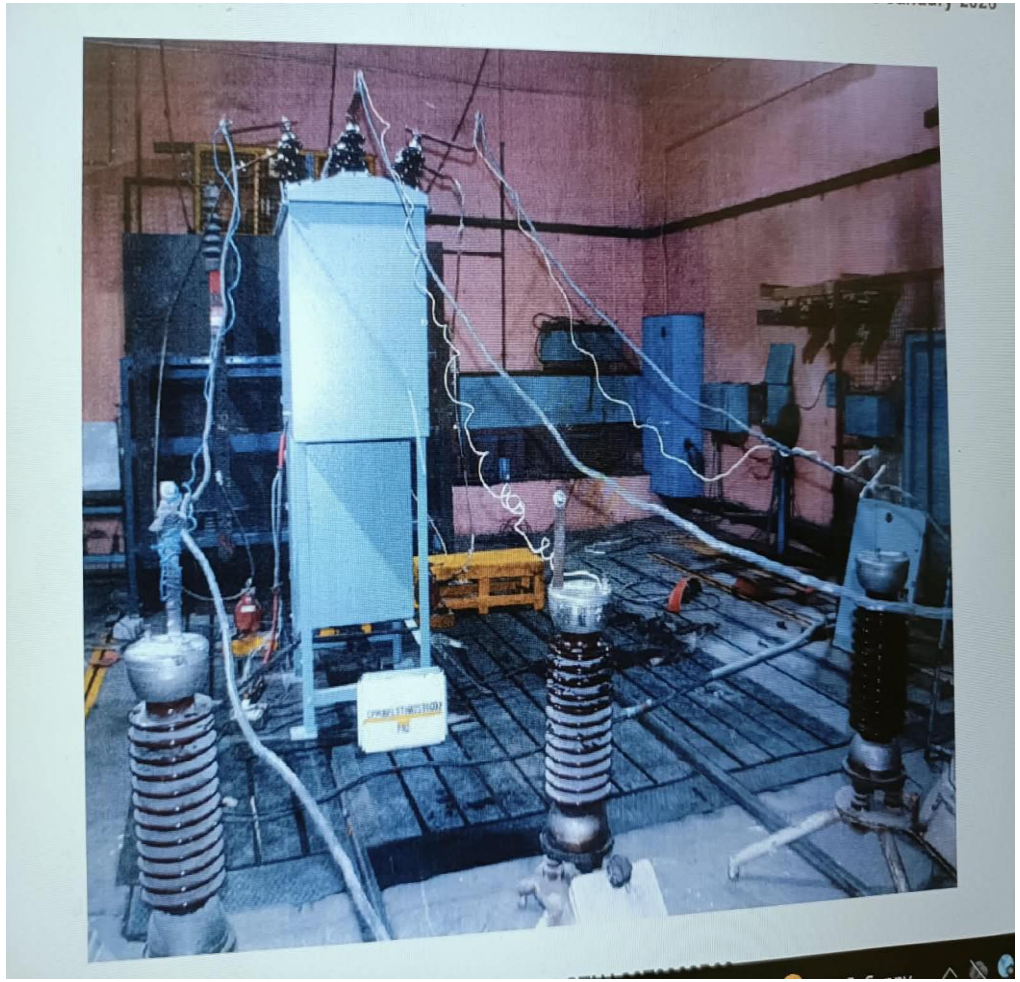
Type Test lab– KERI in 2026:- 36 KV VCB Kiosk & 36 KV 31.5 kA PCVCB for E2C2





2a.Details Of Tests

Type Test lab– CPRI: 12 KV VCB Panel /VCB Kiosk for E2C2





2a. Approved laboratory For Type Test of Switchgears

The Short-Circuit Testing Liaison (STL)

Members of STL are:

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For ASTA Intertek
see information below

CESI, Italy

CPRI, India

ESEF ASEFA, France

JSTC, Japan

KEMA, The Netherlands

KERI, South Korea

PEHLA, Germany

SATS, Norway

STLNA, USA

VEIKI, Hungary

ZKU, Czech Republic

In India NABL approved Lab ERDA at Vadodara is also carrying out some basic Type Tests for MV switchgear. CPRI has both in Bhopal & Bangalore are carrying out the Type Tests as per IEC standard.

3.0. Switchgear – VCB- Introduction

VCB stands for Vacuum Circuit Breaker, a type of electrical switchgear used to protect high-voltage systems (typically 11 kV to 33 kV) by interrupting current flow during faults. It uses a vacuum as the medium to extinguish the arc, making it highly reliable and efficient.

Construction & Working : Main Components:

- **Vacuum Interrupter**: Contains fixed and moving contacts in a sealed chamber.
- **Operating Mechanism**: Opens/closes contacts manually or automatically.
- **Insulating Housing**: Provides mechanical support and insulation.

- **Working Principle**:
  - When a fault occurs, contacts separate inside the vacuum chamber.
  - The arc forms briefly but is extinguished almost instantly at the current zero point.
  - The vacuum prevents ionization, ensuring safe interruption.

3.0. Switchgear – VI Introduction

What is a Vacuum Interrupter?

- A **vacuum interrupter (VI)** is the core switching element used in **medium voltage (1 kV–36 kV)** circuit breakers. It interrupts current by separating contacts inside a **sealed vacuum chamber**, where there is **no medium (like air or oil)** to sustain an arc.

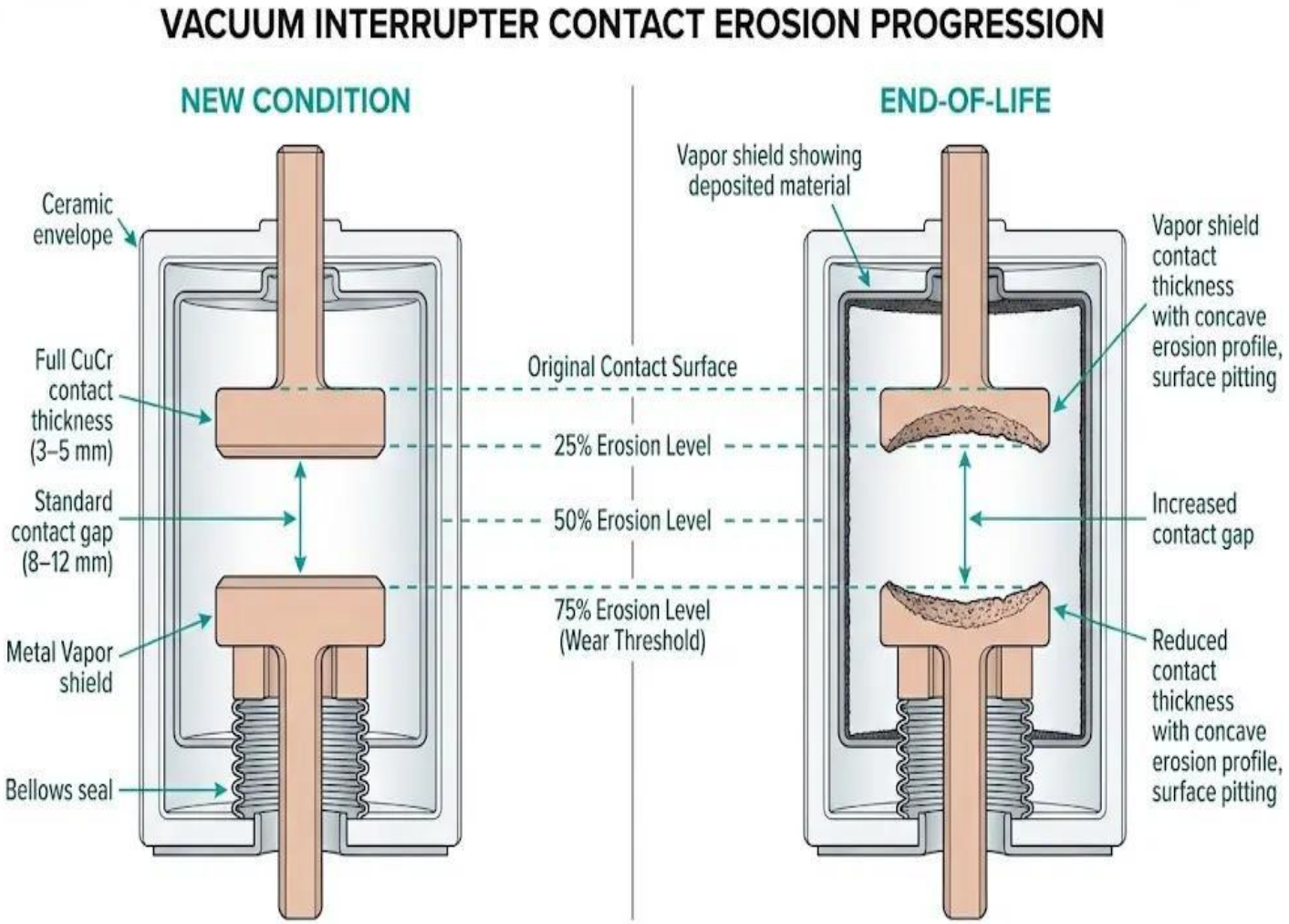
Application of VI

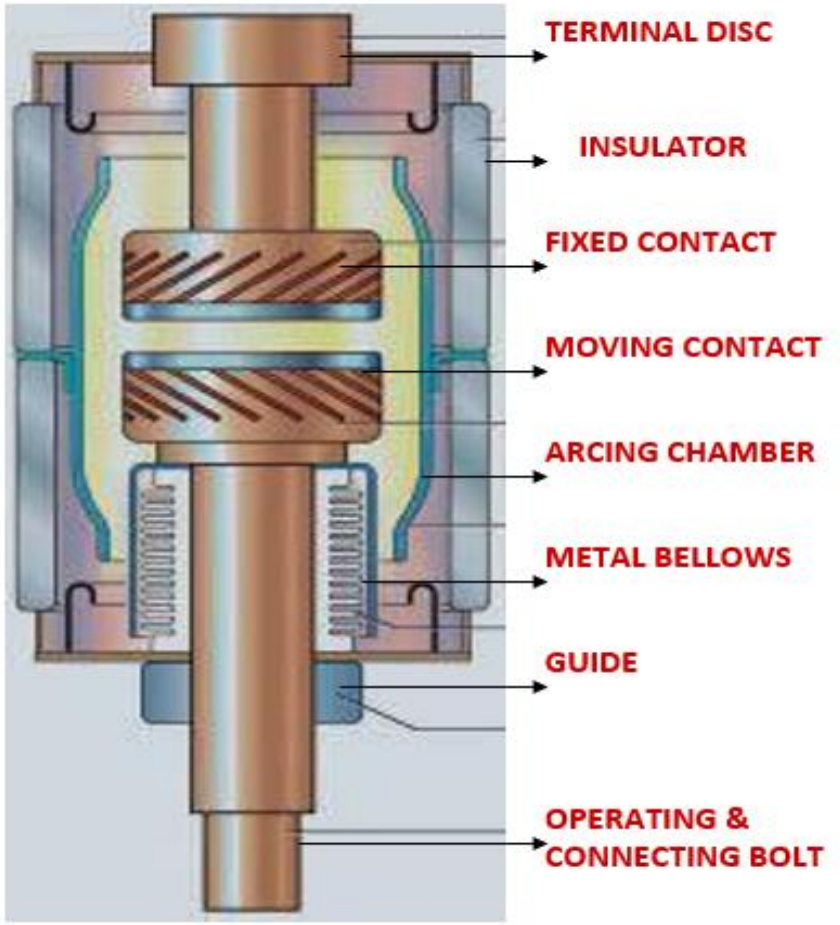
- Vacuum interrupters are primarily used in medium- and high-voltage circuit breakers, load switches, and protective switchgear because they safely extinguish electrical arcs in a compact, reliable, and maintenance-free design. They are widely applied in power transmission, distribution, industrial plants, railways, and arc furnace systems.

3.2 How a Vacuum Circuit Breaker Works

- **Closed state:** touching contacts carry normal current.
- **Opening:** the mechanism separates the contacts and a metal-vapour arc forms.
- **Arc control:** the contact geometry manages the arc and heat distribution.
- **Current zero:** the arc extinguishes and metal vapour condenses.
- **Dielectric recovery:** the contact gap rapidly regains insulation strength.

3.0 Switchgear – VI Sectional Views





3.0 Switchgear – VI Details

Contacts Design

RMF (Radial Magnetic Field) and AMF (Axial Magnetic Field) contacts are two arc-control techniques used in vacuum circuit breakers **(VCBs).**

RMF rotates the arc along the contact surface to spread heat and reduce erosion, while **AMF** stabilizes the arc in a diffuse mode across the contact gap, minimizing localized damage.

Key Insights:

RMF Contacts: Best for moderate current levels, simpler design, and effective arc rotation. However, at very high currents, arc instability (pinching) can occur.

AMF Contacts: Superior for high-current interruption, as the diffuse arc minimizes erosion and ensures reliable performance. They require more complex design and precise orientation.

Type of Contact Design

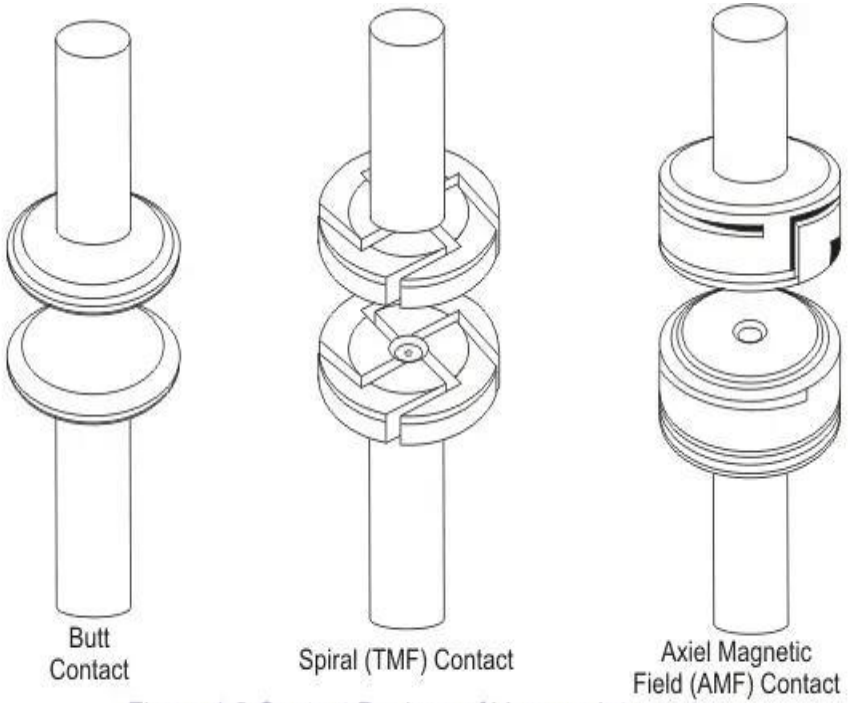


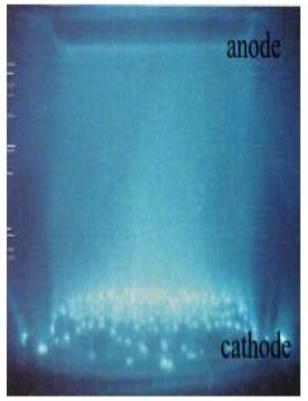
Figure 1.2 Contact Designs of Vacuum Interrupter

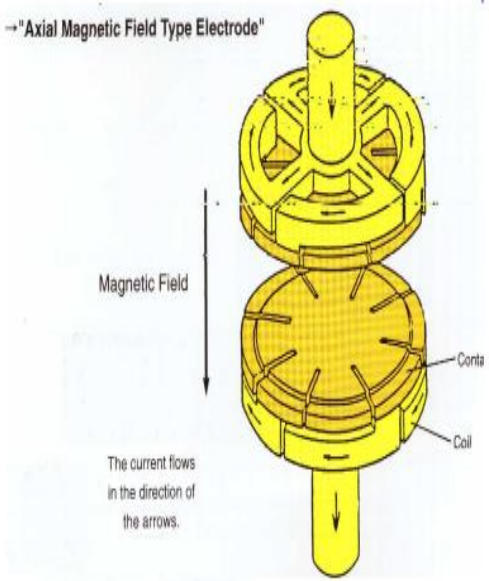
A. Radial Magnetic Field (RMF) geometry Figure shows the contacts tested. These are of the “Swastika” type and create the radial magnetic field by forcing the current to flow along the “petals” formed by the slotting of the contact disc.

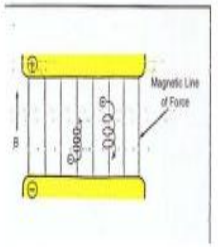


Arc Control Geometry- Axial Magnetic Field - AMF

- arc is diffused by the action of axial magnetic field generated by the current itself
- arc occupies the whole space between the electrodes uniformly



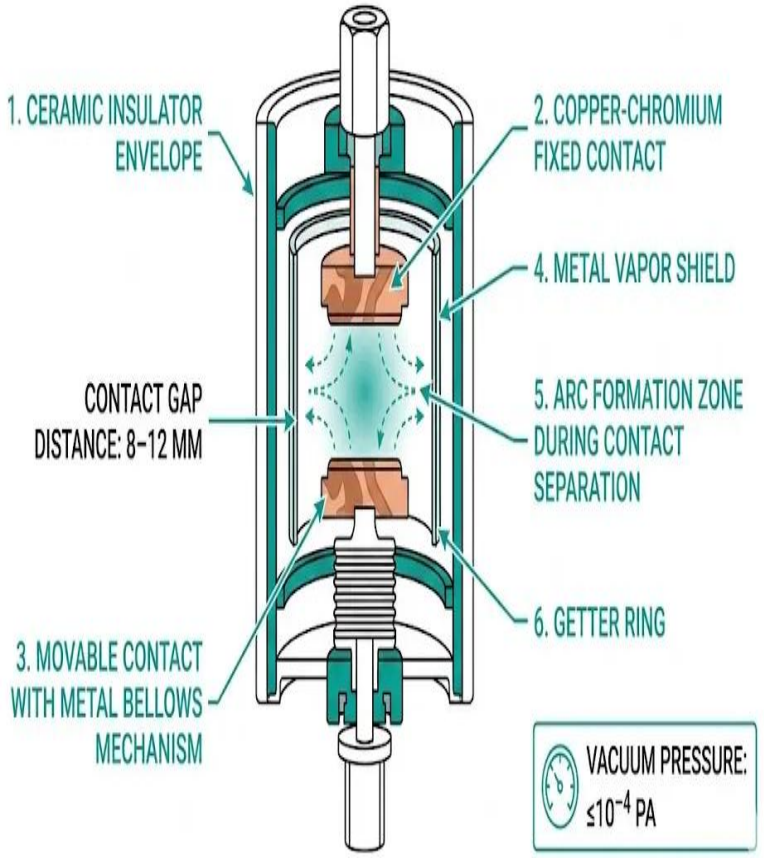




3.0 Switchgear- VI Technology Overview

Core Technical Elements:

- **Hermetic Deep-Vacuum Chamber:** Operates within an ultra-clean environment sealed at an internal pressure of **(10^{-4}Torr) to (10^{-6} Torr)**, utilizing vacuum as the ultimate dielectric insulation medium.
- **Dynamic Arc Control Physics:** When contact faces separate under electrical load, a local arc burns inside evaporated metal vapor. At the AC current zero point, vapor production stops instantly, causing rapid dielectric recovery and successful interruption.
- **Advanced Material Engineering:** Utilizes specialized **Copper-Chromium (Cu-Cr)** alloy contacts optimized for transverse or axial magnetic field control to actively diffuse current distribution and minimize surface wear.
- **Mechanical Flex-Life Integrity:** High-purity stainless steel bellows absorb high-impulse linear driving motion while permanently retaining hermetic sealing over a 20-year lifecycle.



3.0 Switchgear- VI Technology Overview

Strategic Value Matrix of In-House VI Facilities

- **Supply Chain Sovereignty:** Eliminates single-vendor dependencies and logistics overheads on critical, long-lead sub-components, directly matching furnace output directly with breaker factory assembly lines.
- **Strict Process Compliance (ISO 9001 Guidelines):** Direct environmental monitoring of particulate levels inside strict **Class 1,000 / Class 10,000 cleanrooms** to reduce internal contamination risks.
- **Advanced Thermal Optimization:** Precise tracking of specialized multi-stage vacuum furnace parameters, ensuring clean outgassing holds at **350°C, 550°C, and 750°C** before triggering liquidus braze reflow.
- **Floor-Level Testing Rigor:** Direct deployment of high-pressure helium bombing vessels and mass-spectrometer leak diagnostics.

Vacuum Interrupter(VI)

1 SVI1226-16A
(12kV 1600A 26.3kA)



2 SVI1226-20A
(12kV 2000A 26.3kA)



3 SVI3625-16A
(36kV 1600A 26.3kA)



4 SVI1221-06A
(12kV 630A 21kA)



VI Assembly Process



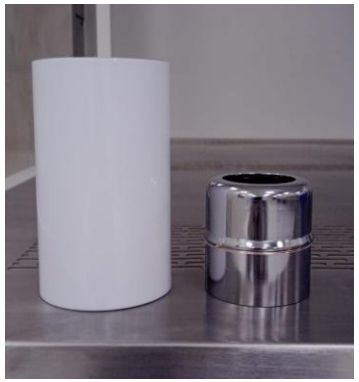
VI Parts



Fixed Sub Assy.



Moving Sub Assy.



Ceramic Shield



Final Assy.

Vacuum Interrupter – Process & Testing







Assembly

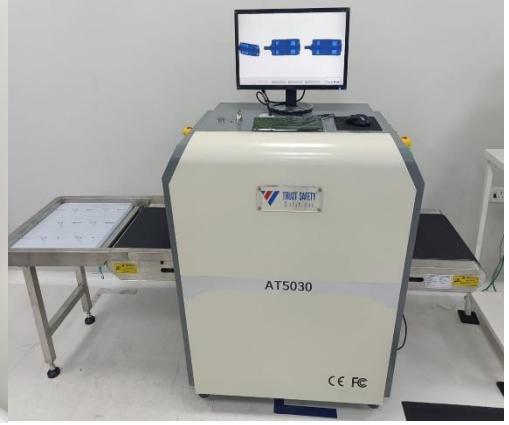


Brazing









Testing

4.0 Digitalization of Rural Substation

IoT-Enabled Retrofit
for Supplied 11/36 kV VCB Panels

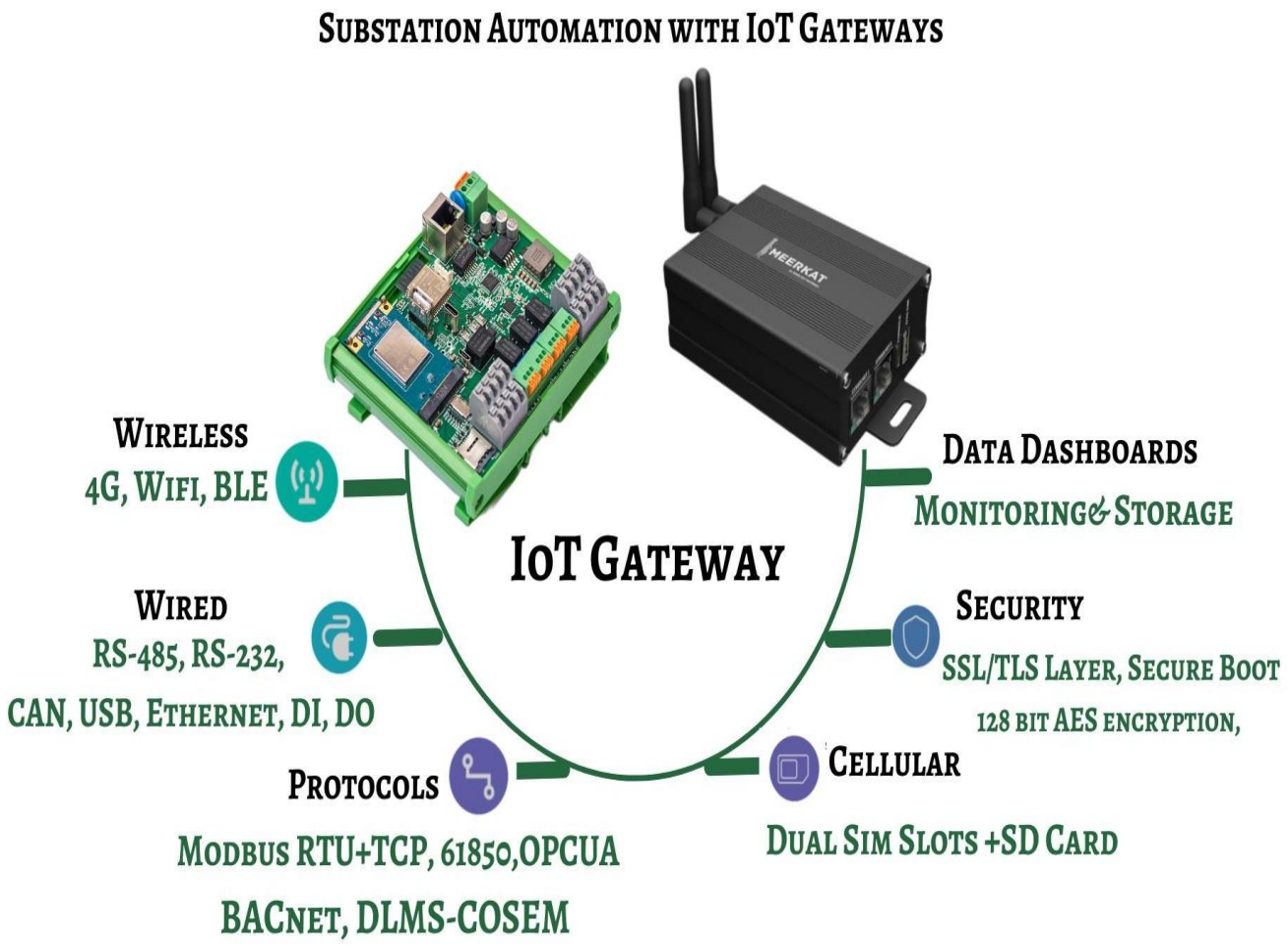
Breaker health • Operation count • Fault Trip, Current, voltage, power & energy & Extended to Transformer

Business model: one-time retrofit and integration charges (CAPEX)

4.1 IoT for Rural Substation Monitoring

- Remote and rural substations need visibility of equipment condition and operating status.
- IoT monitoring brings Circuit breaker, transformer and power-system data to remote users.
- Trends and alarms support fault diagnosis, maintenance planning and outage response.
- Stelmec’ s indoor and outdoor VCB installations provide a basis for adding remote monitoring as established a **cost-effective solution in Rural substation** to facilitate the customer with required Data by remotely. **Digitize the installed panel—without replacing the switchgear, etc.**

4.2 IoT Gateway Features



- Connects relays, IEDs and sensors.
- Interfaces: RS-485, Ethernet, USB, CAN, 4G and Wi-Fi.
- Protocol conversion: Modbus and IEC 61850 to MQTT.
- RS-232 connects compatible panel meters.
- Cloud telemetry and SD-card storage support store-and-forward.
- Confirm encryption, TLS and remote-control permissions.

4.0 IOT GATEWAY FEATURES

Basic Working Principle

1. Data Acquisition : From CB, Transformer, etc.
2. Local Processing : IEDs & Sensor data
3. Communication : Data is transmitted through:
   - IEC 61850 communication protocol
   - Ethernet/fiber-optic networks
   - Wireless technologies (4G/5G, LoRa, NB-IoT)
4. Cloud/Edge Analytics
5. Remote Monitoring & Control

Key Components of IoT Substation Monitoring Solutions

1. Sensor Networks
2. Communication Infrastructure
3. Edge Computing
4. Cloud-Based Platforms
5. Predictive Analytics

Scalability and Interoperability

4.0 IOT – Substation & Automation Data

Data's from Circuit Breaker side

- **Data from Relay** :RYB Current, Fault data, CB Status, etc.
- Data's from TVM : RYB Current & Voltage& Power Management
- Data from CB panel: Temperature, Humidity, Arc Flash over, Partial Discharge, etc.by adding suitable Sensors
- Special data from VCB : Displacement, Life of the VI, Mechanical Operation (Sensor under development)

Substation Data's

- Transformer Data: RYB current & Voltage, Oil & wind temp, Buchholz trip, etc.
- AQI, GHS, Battery voltage, (Finding Earthling healthiness), etc.

Displayed data at Customer end:

Analyzed Data's On predictive maintenance, Alert on emergency, Life cycle management, SAIDI and SAIFI ,Outage management, etc. & can be AI empowered

4.0 Benefits of IOT Based Substation

Benefits of IoT-Based Substations

- ✅ Real-time monitoring
- ✅ Predictive maintenance
- ✅ Reduced outages & Faster Fault Response
- ✅ Improved reliability
- ✅ Remote operation capability
- ✅ Lower maintenance cost
- ✅ Better asset utilization
- ✅ Enhanced grid automation

If added extended software

SAIDI (System Average Interruption Duration Index)

$$\text{SAIDI} = \frac{\text{Total duration of sustained interruptions}}{\text{Total number of customers}}$$

SAIFI (System Average Interruption Frequency Index)

$$\text{SAIFI} = \frac{\text{Total number of sustained interruptions}}{\text{Total number of customers}}$$

4. 0 IOT Security

Secure Boot Enabled – Physical Security of Fuse Bits on Board

FOTA over HTTPS/MQTTS with links on Utility Server or MEITY Tier 3 Cloud so data remains on Indian servers

Encrypted executable code with device specific random key for code protection at rest

Data Encryption with device specific key

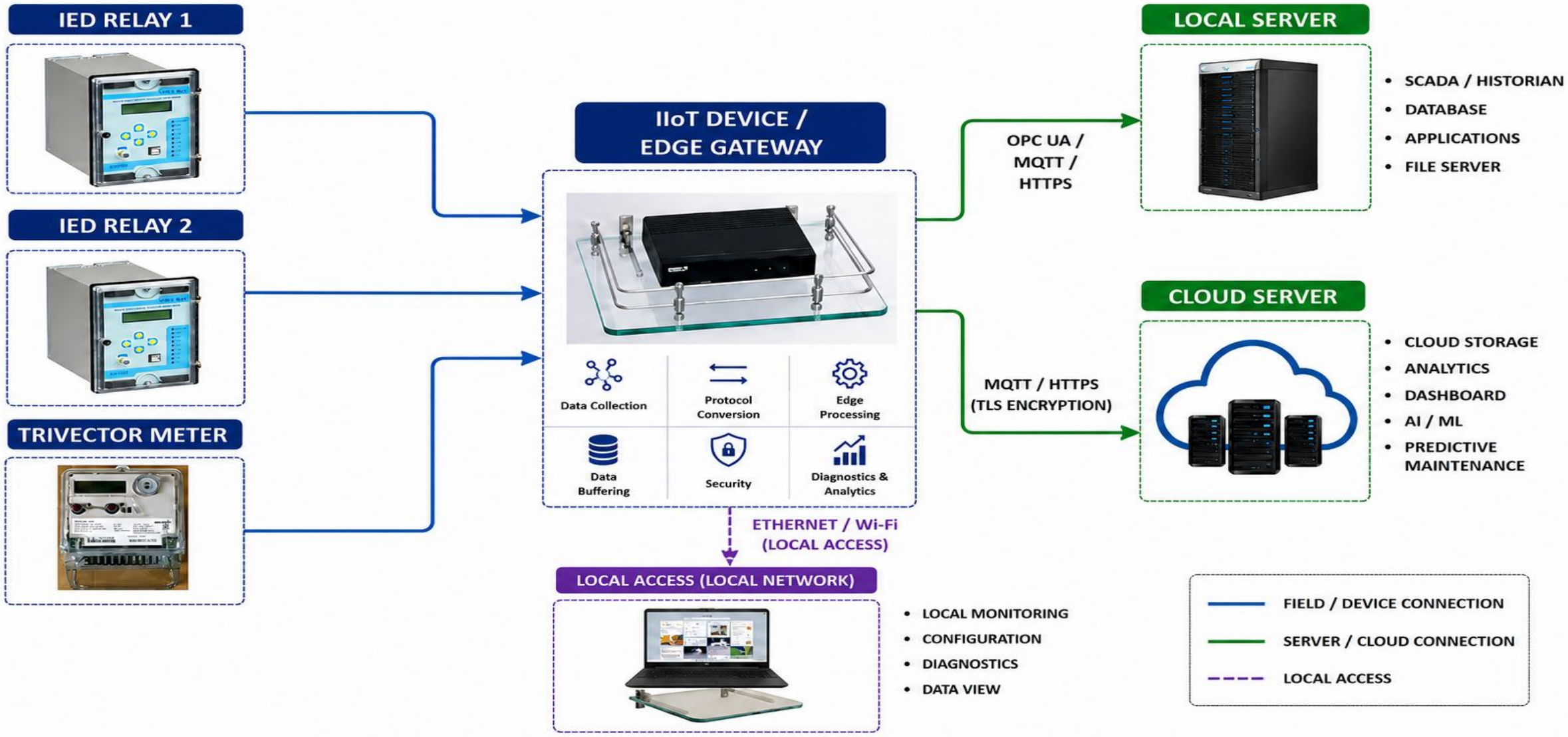
Wireless transmission over HTTPS or MQTTS

Encrypted key store protected against unauthorized read, including side loading of firmware with support for TLS 1.2/1.3

Hob & SBoM for part & supply chain traceability to eliminate backdoor channels in hardware or software

Dual Sim behind board with no physical access

4.0 IOT Substation Architecture Network



5. 0 AI- in Substation & Automation

AI in Substation & Automation

Medium Voltage (MV) substation automation, Artificial Intelligence (AI) refers to the integration of machine learning algorithms and computational techniques to enable smarter, more autonomous, and data-driven management of the substation's operations. This moves beyond traditional, rule-based automation (SCADA) by allowing systems to learn from data, identify complex patterns, and make intelligent decisions in real time to enhance efficiency, reliability, and safety.

Key Functions and Applications AI in MV substation automation leverages vast amounts of data from intelligent electronic devices (IEDs), sensors, and other equipment to enable a range of advanced functions:

5.1 AI- in Substation & Automation

AI vs DMS vs ADMS

(Power Distribution Automation Context)

Simple One-Line Definitions

DMS

Software to monitor, analyse, and optimize distribution networks

ADMS

Integrated platform combining DMS + OMS + SCADA

AI

Intelligence layer that learns from data and supports decisions

👉 AI is not a system like DMS/ADMS – it is an enabler used inside them.

5.1 AI, DMS & ADMS

- **DMS:** software for monitoring, analysing and optimising distribution networks
- **OMS :** Detect outages Identify affected customers Manage outage crews Track restoration progress Calculate reliability indices (SAIDI, SAIFI, CAIDI)
- **ADMS:** an integrated platform combining distribution management, outage management and SCADA capabilities.
- **AI:** analytical methods that learn from data and support decisions.
- **AI** can enhance DMS and ADMS functions, with implementation defined by the utility’s needs.

5.2 AI- in Substation & Automation

Why AI is Needed in Substations?

Traditional substation automation is: Rule-based (fixed settings, thresholds), Reactive (trip after fault) , Data-rich but insight-poor

AI makes substations: Predictive, Self-learning, Condition-aware, Grid-adaptive

Key AI Use Cases in Substation Automation

1. Predictive Maintenance (BIGGEST USE CASE) : Equipment covered:

Power transformers, MV/HV switchgear (VCB, GIS), CT/PT, Cables & joints,

Surge arresters, etc.

AI inputs: Partial discharge (UHF / TEV / HFCT),Thermal data, SF_6 density & moisture , Vibration & acoustic data, Breaker operation counters, etc

AI output : Remaining Useful Life (RUL),Failure probability, Maintenance prioritization ,etc..

5.3 AI- in Substation & Automation

AI Architecture in Substation Automation Copy code

Sensors / IEDs / IoT Devices

↓

Bay Level / Process Level

↓

Edge AI (Gateway / Substation Server)

↓

Central AI Platform (Utility Cloud / Data Center)

↓

SCADA / ADMS / EMS

Where AI Runs : Level AI Role Edge, Fast analytics, anomaly detection, Substation Server, Asset health, PD classification, Cloud, Fleet analytics, learning models

5.4 AI- in Substation & Automation

AI + IoT Device Examples (MV Focus)

Equipment: IoT + AI Function VCB, Mechanism health, contact erosion , GIS SF_6 leakage prediction ,Transformer, Hot-spot & ageing ,Cables ,Joint failure Prediction ,Panels Internal PD detection , etc.

Protocol Integration (Very Important) : Layer ,Protocol, IED ↔ Gateway

IEC 61850 (MMS, GOOSE, SV) ,IoT Devices, MQTT / OPC UA / Modbus ,

Gateway ↔ SCADA ,IEC 60870-5-104 / DNP3 ,AI Platform ↔ SCADA

IEC 104 / REST API

Predictive Maintenance Alarms Example:

“VCB feeder-3: Contact wear 78%, estimated failure in 6 months”

Digital substations + AI = *Predict failures, act fast & Optimize in real-time.*

Save money, enhance safety & - Transform into smart, self-managing assets

5.5 AI- in Substation & Automation

Digital substations empowered with AI are game-changers! Top advantages:

1. **Predictive Maintenance & Reliability-**

Proactive problem-solving: AI analyzes sensor data (vibration, temperature, oil quality) to predict equipment failures before they happen.-

Reduced downtime:

Fix issues before they impact power supply. Less unplanned outages = happier customers!- Example: Detect a transformer anomaly and schedule maintenance saving millions**.**

2. Real-Time Monitoring & Faster Response- Instant anomaly detection: AI continuously monitors voltages, currents, and grid conditions. If something's off, alerts are triggered instantly.- Self-healing grids: AI can automatically isolate faults, reroute power, and restore service — all in milliseconds.- Result: Improved SAIDI (System Average Interruption Duration Index).

5.6 AI- in Substation & Automation

3. Optimized Operations & Efficiency- Load forecasting and energy optimization: AI predicts consumption patterns, adjusts tap changers, and manages reactive power — reducing losses.- Automated decision-making: No more manual tweaks! AI balances the grid dynamically, improving efficiency.- Less human intervention:

4. Enhanced Safety & Security- Cybersecurity protection:

AI monitors substation networks for intrusions, anomalies, or attacks. Protects critical infrastructure.- Risk mitigation: Identify abnormal conditions (like overloading or arc risks) and act fast.- Safer operations for humans and equipment.

5. Data-Driven Insights & Planning- Turns data into gold:

AI analyzes decades of operational data, weather, load patterns, and equipment health.- Helps plan upgrades, anticipate future needs, and make substations “future-proof.”

5.7 AI- in Substation & Automation.

6. Cost Savings (the big one!)-

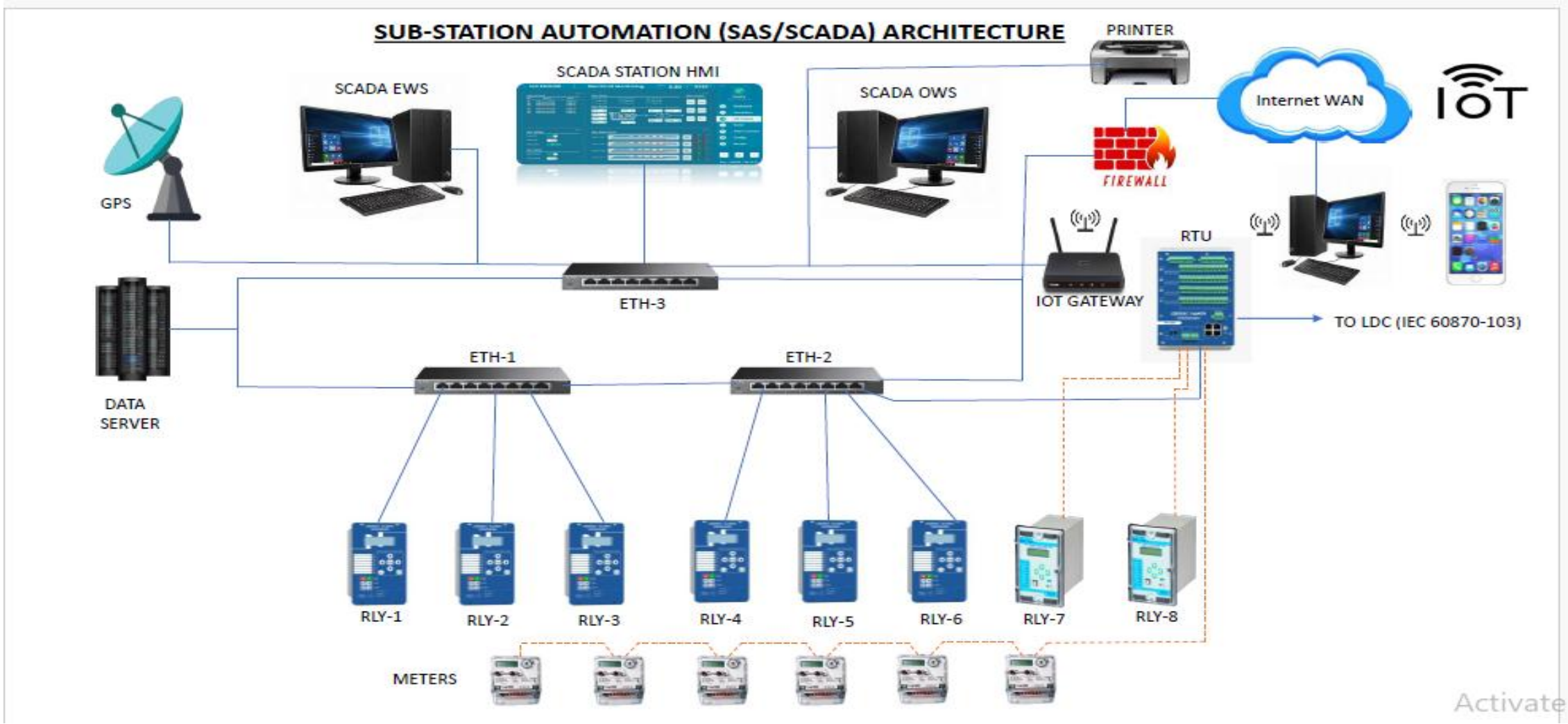
Fewer outages = fewer penalties and repairs.- Efficient resource use: Optimal transformer tap settings, reduced losses, and smart asset management.- Labor savings: Automation reduces the need for on-site staff, especially in remote locations.- Longer equipment life: Predict and prevent failures.

7. Integration with the Smart Grid-

AI turns substations into intelligent nodes of a modern grid:- Responds to fluctuating renewables (solar, wind).- Participates in demand response, helping stabilize the entire ecosystem.- Enables peer-to-peer energy trading and microgrid management.

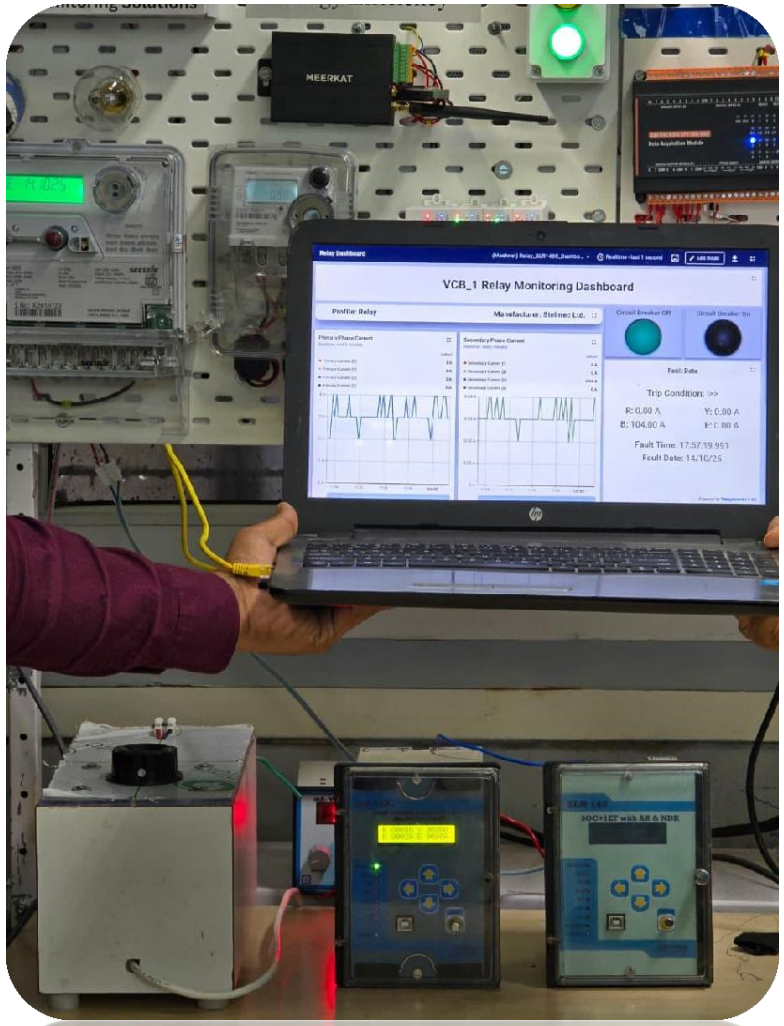
8. Scalability & Flexibility- Deploy the same AI "brain" across hundreds of substations. Centralized learning, local execution.- Adapts as needs evolve — new sensors, new conditions, new efficiencies.

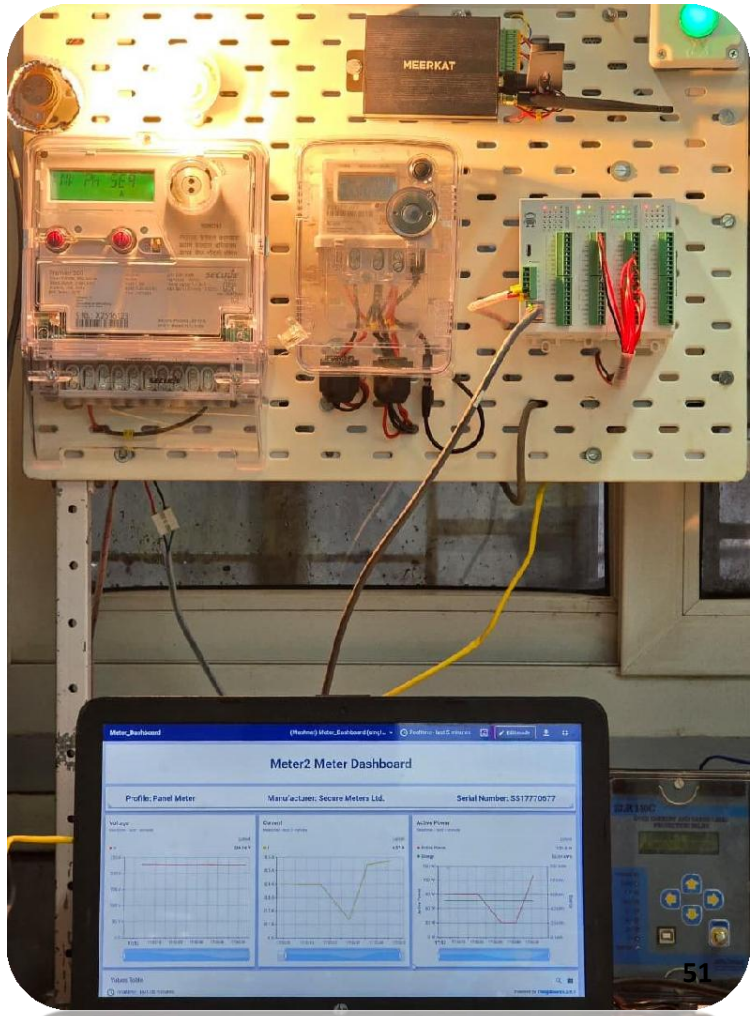
5.0 Network


SUB-STATION AUTOMATION (SAS/SCADA) ARCHITECTURE
PRINTER
SCADA STATION HMI
SCADA EWS
SCADA OWS
Internet WAN
IoT
GPS
FIREWALL
RTU
IOT GATEWAY
ETH-3
TO LDC (IEC 60870-103)
ETH-1
ETH-2
DATA SERVER
RLY-1
RLY-2
RLY-3
RLY-4
RLY-5
RLY-6
RLY-7
RLY-8
METERS

6.0 Case Studies – Indoor VCB



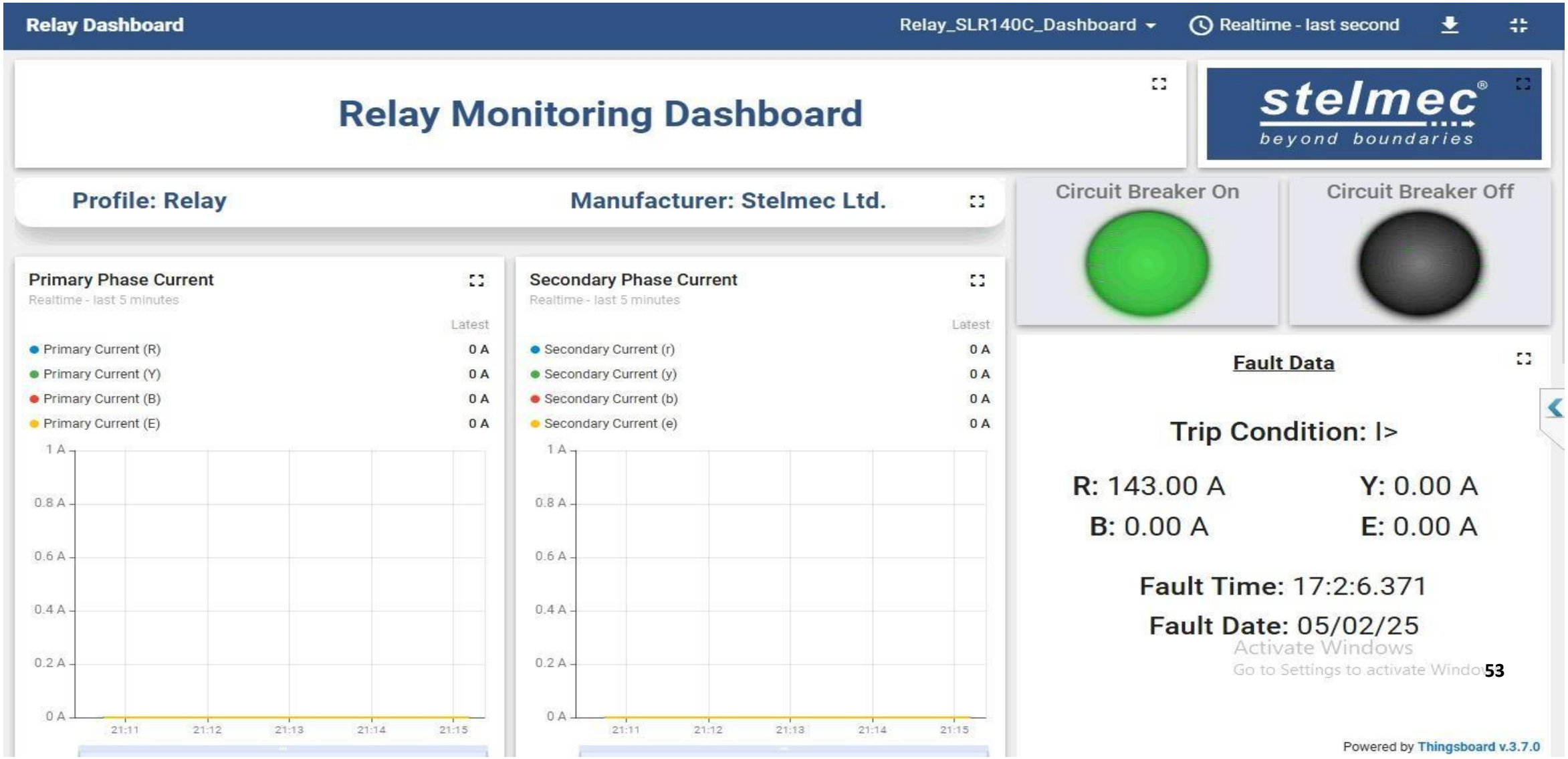




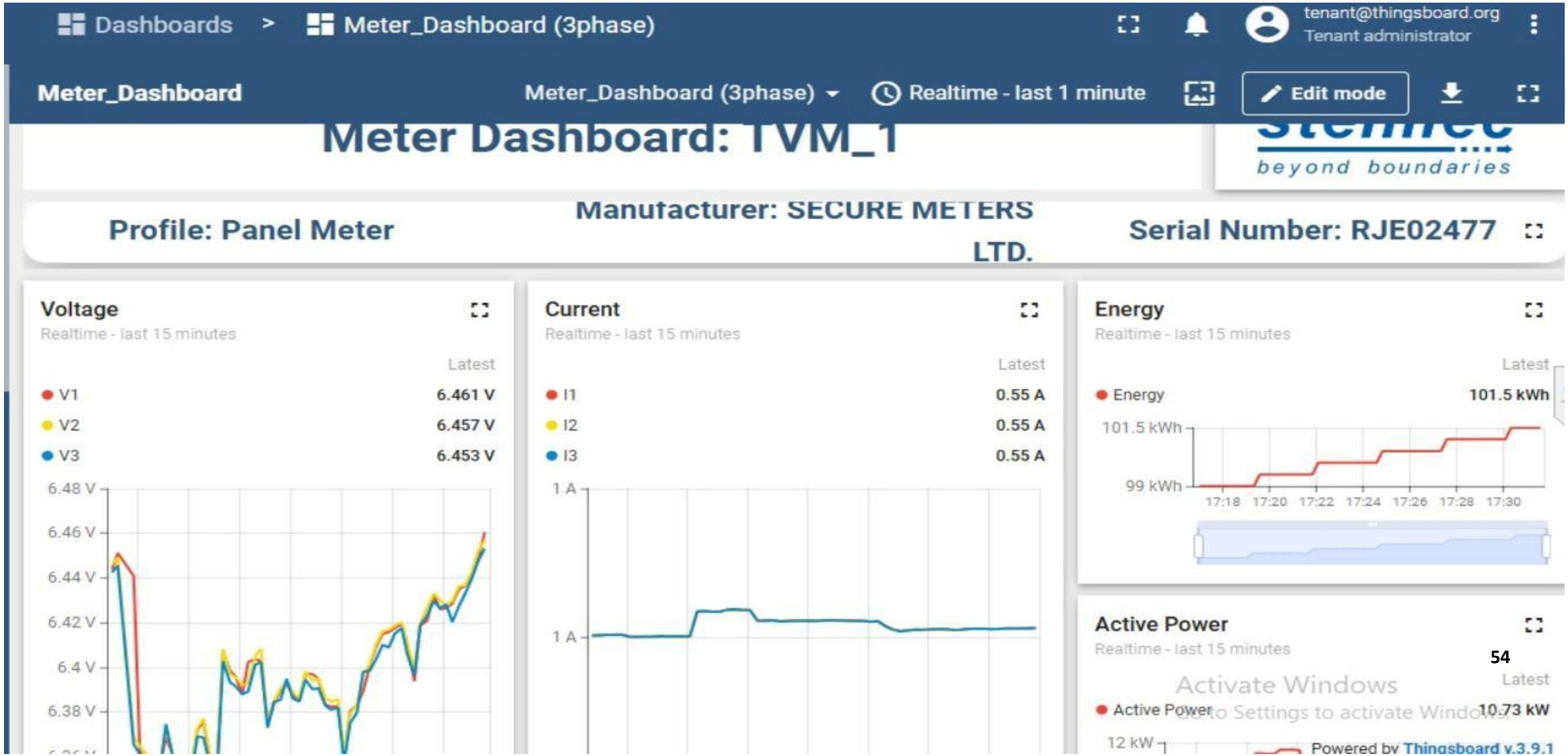
6.1 Case Studies – Outdoor VCB



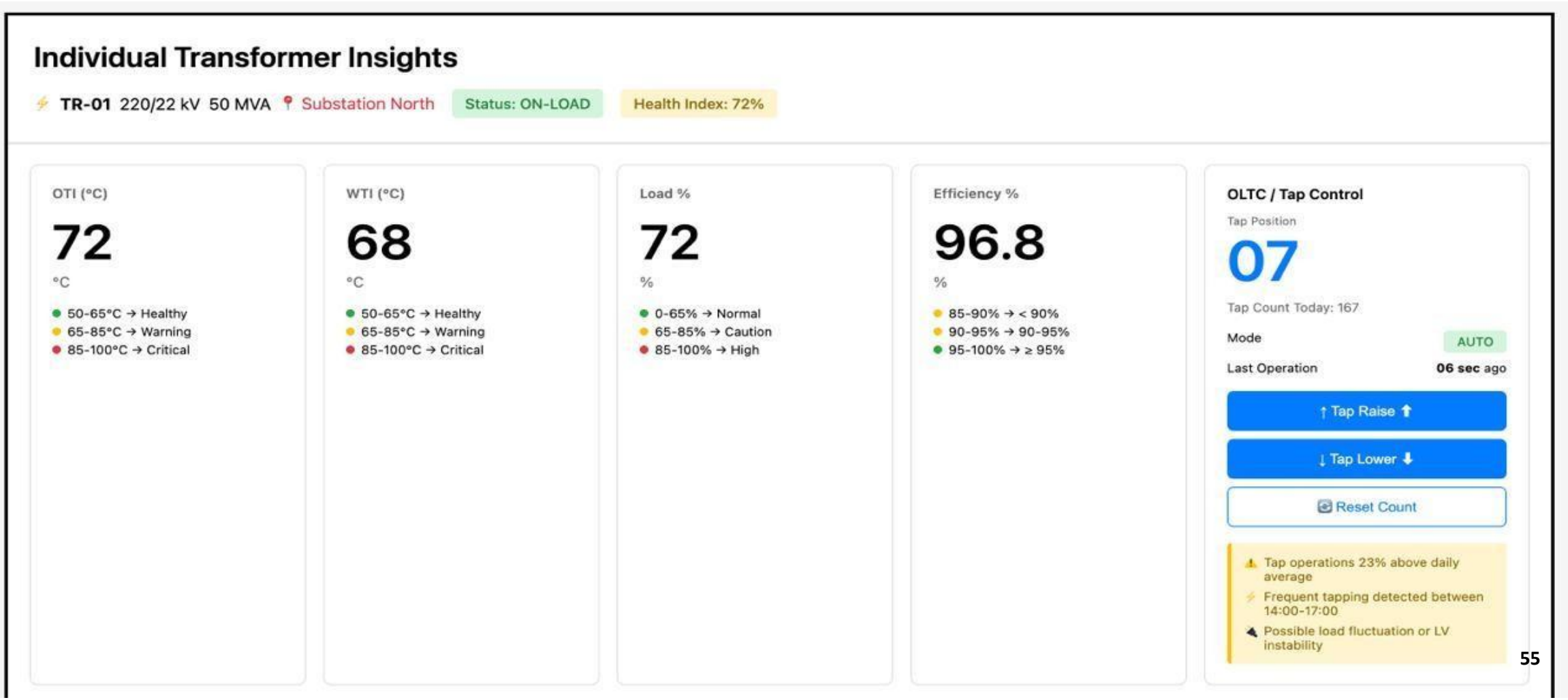
6.2 IoT - Dashboard



6.3 IOT GATEWAY Dashboard



6.4 Case Study - Transformer Monitoring



6.5 Case Study- Transformer Monitoring

Primary Side (HV) - 220 kV

VOLTAGES PER PHASE

Phase A	Phase B	Phase C
127 kV	127 kV	127 kV

CURRENTS PER PHASE

Phase A	Phase B	Phase C
420 A	415 A	425 A

ADDITIONAL PARAMETERS

Earth Leakage	Frequency
2.8 mA	50.4 Hz

Secondary Side (LV) - 22 kV

VOLTAGES PER PHASE

Phase A	Phase B	Phase C
11.6 kV	11.5 kV	13.1 kV

CURRENTS PER PHASE

Phase A	Phase B	Phase C
4150 A	4100 A	4200 A

ADDITIONAL PARAMETERS

Earth Leakage	Frequency
2.0 mA	49.8 Hz

Winding Temperatures

PRIMARY WINDINGS

Phase A	Phase B	Phase C
67.6°C Healthy	69.4°C Healthy	68.9°C Healthy

SECONDARY WINDINGS

Phase A	Phase B	Phase C
71.2°C Warning	70.5°C Warning	72.5°C Warning

6.6 Case Study- Transformer Monitoring

Oil Level + Moisture



Moisture (ppm)

26.8

Trend: ↑

Minimum Oil Level

- Moisture increased 6ppm in 12 hours
- Approaching 30ppm warning limit

Dissolved Gas Analysis (DGA)

H_2 Gas **145** ppm	CH_4 Gas **27.8** ppm
C_2H_6 Gas **11.4** ppm	CO Gas **340** ppm
CO_2 Gas **2800** ppm	

- H_2 rising steadily for 48 mins
- Possible partial discharge or insulation stress
- Monitor closely

Protection System Status

Buchholz Relay	✓ Normal
PRD	✓ Normal
OLTC Surge	✓ Normal
OTI Trip	✓ Normal
WTI Trip	✓ Normal

- ✓ No protection trips in last 24 hours
- ✓ System operating stable

6.7 Case Study- Transformer Monitoring

Voltage Stability

LV fluctuating in 90-110% band
2 out-of-band dips detected

Load Stress

Avg load 72% for 6 hours
Peak load 79% | 11% higher than fleet average

Temperature Duration

WTI > 70°C for 3.2 hours today
Max: 74°C | Limit: 85°C

Additional Metrics

Total Energy: 1,247 MWh Today

Uptime: 99.8% This Month

Peak Load: 79% @ 15:30

Active Alerts: 3 Current

Last Maintenance: 15 days ago

Next Service Due: In 45 days

7.0 Conclusion & Next Steps

- **Reliable rural substations combine qualified switchgear with useful equipment data.**
- **In-house VI capability supports product development and manufacturing control.**
- **IoT monitoring provides visibility for diagnosis and maintenance decisions.**
- **AI can support condition assessment after data and model validation.**
- **Proposed next step: agree a pilot scope, security controls and measurable success criteria.**

7.0 Conclusion & Next Steps

The proposed framework provides rural utilities with an affordable pathway to remote visibility and condition-based maintenance where full SCADA deployment is not economically viable. A representative use case is a rural 33/11 kV substation in which transformer, VCB, CT/PT, battery, and temperature data are collected at the edge, transmitted through low-cost IoT networks, and analyzed to generate an Asset Health Index and maintenance warnings. The modular architecture can also be extended to distribution transformers and integrated with utility SCADA or advanced distribution management systems (ADMS).

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THANK YOU

Questions & Open Discussion

R S Selvam | VP – R&D |
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Education Series